
TD3 FOR LEARNING TO WALK

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ABSTRACT

This document provides an overview of two RL implementations based on the course notebooks uploaded: a base agent for the `rldurham/Walker` task and an expanded variant for Walker-Hardcore. Both agents use the framework of Twin Delayed Deep Deterministic Policy Gradient (TD3) [1] with continuous actions, target policy smoothing, delayed actor updates, Polyak averaging, and replay-based off-policy learning. The standard Walker agent employs two-layer 256-unit networks, a 2,000-step warm-up phase, and decaying Gaussian exploration. The hardcore agent uses wider 400–300 networks, a preallocated replay buffer, improved exploration, and a warm-up period of 10,000 steps; it also scales rewards and clips gradients. From an empirical perspective, the Walker agent achieves stable returns greater than 200 during the last phase of training, reaching a best episode return of 211.96 and a mean return of 208.13 over the final 50 episodes. The hardcore version shows a large improvement during training but remains unsolved, ending with a mean return of 72.39 and a best return of 178.74 over the final 50 episodes. These results demonstrate that a tuned TD3 pipeline is sufficient for the base task, whereas the harder terrain requires better exploration, more robust credit assignment, or additional algorithmic modifications.

1 METHODOLOGY

Implementation of Twin Delayed Deep Deterministic Policy Gradient (TD3) [1], which builds on Deep Deterministic Policy Gradient (DDPG) [2], in a walking environment. The action space consists of four continuous actuator commands constrained to the range $[-1, 1]$, and the observation space contains 24 continuous features. The tanh layer at the output of the actor ensures bounded actions, and two Q -functions are used in the critic to reduce positive bias through temporal-difference updates.

For the base Walker notebook, we use a compact architecture with two hidden layers of 256 units in both actor and critic. In the Walker-Hardcore notebook, the network width is increased to 400 and 300 units to handle the more challenging environment containing obstacles.

The policy selects an action at interaction step t as

$$a_t = \pi_\theta(s_t) + \epsilon$$

where ϵ is Gaussian exploration noise truncated to remain within the valid action range. After collecting a small amount of experience, both agents transition from uniform random exploration to policy-driven actions with noise. Transitions $(s_t, a_t, r_t, s_{t+1}, d_t)$ are stored in replay memory and sampled as mini-batches to optimize the critics, following standard reinforcement learning practice [3].

The TD3 target used in both notebooks is

$$y = r + \gamma(1 - d) \min(Q'_1(s', a'), Q'_2(s', a'))$$

$$a' = \text{clip}(\pi'(s') + \epsilon_{\text{targ}}, -1, 1)$$

$$\epsilon_{\text{targ}} \sim \text{clip}(\mathcal{N}(0, \sigma^2), -c, c)$$

The critics are trained per environment step using mean-squared Bellman error, while the actor is updated every two critic updates. Polyak averaging with $\tau = 0.005$ is used to update target networks.

The main design differences between the two notebooks are practical rather than conceptual. The Walker configuration uses `reward_scale = 1.0`, `warmup_steps = 2000`, and episode-based noise decay from 0.1 to 0.01. In contrast, Walker-Hardcore uses `reward_scale = 5.0`, `warmup_steps = 10 000`, slower step-based decay from 0.2 to 0.01, a NumPy-backed replay buffer for faster storage, and gradient clipping applied to both actor and critic networks.

Setting	Walker	Walker-Hardcore
Episodes	1,000	2,000
Hidden layers	256–256	400–300
Warm-up steps	2,000	10,000
Exploration noise start	0.10	0.20
Reward scaling	1.0	5.0
Replay buffer	deque	preallocated NumPy
Gradient clipping	No	Yes (norm 1.0)

Table 1: Main experimental settings for the two notebooks.

2 CONVERGENCE RESULTS

The smoothed training return for both runs is compared in Figure 1. The regular Walker agent exhibits the expected TD3 learning curve: strongly negative early episodes, a swift transition to positive reward, and late-stage performance above the 200 return range. Quantitatively, the first 50 episodes have a mean of -126.89 , while the final 50 episodes average 208.13. The highest single-episode return is 211.96 at episode 979. These results indicate that the base notebook not only improves during training but also converges to a stable walking policy.

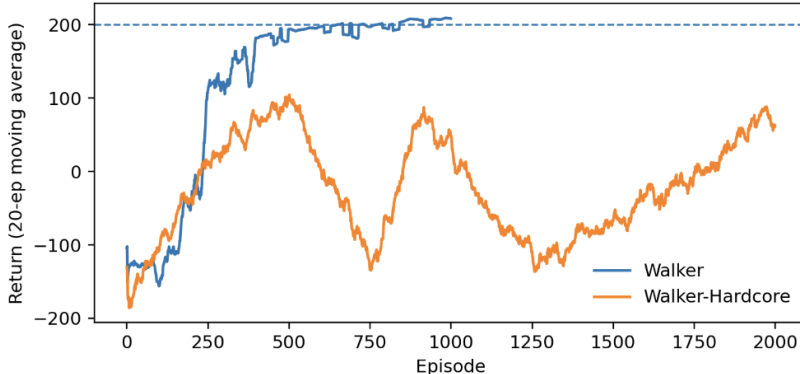


Figure 1: 20-episode moving-average returns for Walker and Walker-Hardcore.

The hardcore task is significantly more difficult. The first 50 episodes average -152.45 , and the learning curve exhibits several cycles of improvement and collapse rather than the smoother monotonic improvement observed in Walker. Nevertheless, the agent learns non-trivial locomotion: the final 50-episode mean reaches 72.39, and the best episode return is 178.74 at episode 473. The final phase remains positive but volatile, suggesting that the policy has learned basic obstacle handling and forward motion but lacks the robustness required to reliably solve the terrain.

Task	First 50 Mean	Final 50 Mean	Peak Return (Episode)
Walker	-126.89	208.13	211.96 (979)
Walker-Hardcore	-152.45	72.39	178.74 (473)

Table 2: Summary of training outcomes measured from the saved logs.

3 LIMITATIONS

The uploaded work is, however, limited in three areas. First, each setting is reduced to a single seeded run, and thus the paper cannot make claims of statistical robustness with respect to random initializations. Second, while the hardcore notebook provides better control over engineering details, it does not introduce a fundamentally new exploration or representation-learning mechanism, which helps explain why learning remains unstable. Third, although evaluation is performed on an independently sampled dataset, the analysis relies on training logs rather than a separate (potentially deterministic) test protocol. As a result, the reported returns reflect both policy quality and residual action noise or training variance.

4 FUTURE WORK

Future work should focus on improving robustness and stability rather than simply extending training duration. Promising directions include running multiple evaluation seeds, increasing the replay warm-up period with prioritized or stratified sampling, and introducing a schedule that reduces target noise once convergence begins. Curriculum transfer from Walker to Walker-Hardcore may also help stabilize learning. Finally, it would be valuable to compare TD3 with Soft Actor-Critic (SAC) under the same logging and computational budget, as SAC may provide stronger exploration on the harder terrain even if the base environment is already solved with TD3.

REFERENCES

- [1] Scott Fujimoto, Herke Hoof, and David Meger. “Addressing function approximation error in actor-critic methods”. In: *International Conference on Machine Learning*. PMLR, 2018, pp. 1587–1596.
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